Ayush Arya Kashyap

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EDUCATION

International Institute of Information Technology, Bangalore **2022 – 2027**
Integrated BTech and MTech in Computer Science Engineering *Bengaluru, India*

EXPERIENCE

Singapore University of Technology and Design (SUTD) **September 2026 – Present**
Visiting Researcher *Singapore*

- Conducting research on LLM-guided fuzzing for AI supply-chain security, targeting vulnerability discovery in serialized machine-learning model formats (e.g., Pickle) and malicious model artifacts distributed through public model hubs.
- Reproduced and benchmarked state-of-the-art systems including ShadowPickle, PickleFuzzer, and DynaHug/DynaOrg in containerized Docker environments, establishing baselines for detecting, analyzing, and fuzzing malicious serialized models.
- Designing and evaluating LLM-guided fuzzing techniques, including semantic-aware mutation, LLM-assisted constraint solving, context construction, and reasoning-based test generation, to reach deeper program states that conventional coverage-guided fuzzers miss.
- Developing context-aware fuzzing pipelines that combine program and model-format semantics with LLM-generated guidance to improve mutation quality and uncover deeper vulnerabilities in binary and serialized model formats.
- Built reproducible experimental pipelines and benchmarks to compare existing fuzzing and model-security approaches on detection coverage, vulnerability discovery, and efficiency.

Microsoft **May 2026 – July 2026**
Cloud Network Engineering Intern *Hyderabad, India*

- Spearheading the development of Rakshak, an AI-driven network change assurance agent for WAN operations, architecting the platform from the ground up to improve network visibility, safety, and operational decision-making.
- Architecting agentic workflows using LLMs and MCP-based tool orchestration to automate pre-change validation, risk assessment, and network change analysis across the change lifecycle.
- Building custom MCP tools for telemetry retrieval, network state analysis, anomaly detection, baseline comparison, and post-change verification using operational network data.
- Engineering context-aware reasoning pipelines that synthesize network logs, telemetry, configuration changes, and operational metadata into actionable insights and human-readable summaries.
- Designing AI-assisted monitoring and impact assessment workflows to detect deviations, evaluate change outcomes, and support large-scale network reliability and visibility initiatives.

Trimble Inc. **June 2025 – August 2025**
Graduate Technical Intern *Chennai, India*

- Developed LedgerScan, an AI-powered web application for automated invoice and receipt data extraction, deployed and actively used by the accounts team for financial document processing.
- Built Trimble Media Coverage Tracker using AI-driven content parsing, classification, and automated Excel reporting; deployed for internal media monitoring and analysis workflows.
- Created an Agent-Powered API Testing Framework using LLMs and Playwright, enabling schema-aware, domain-adaptive, and automated API validation for production applications.
- Implemented GCP-based automation pipelines for invoice processing, data extraction, and enterprise workflow orchestration, with solutions deployed to production environments.
- Delivered end-to-end full-stack features spanning frontend development, backend APIs, AI integrations, and cloud pipelines, with multiple solutions adopted in production workflows.

E-Health Research Center (EHRC) **May 2025 – May 2026**
Research Intern *Bengaluru, India*

- Conducted large-scale neuroimaging analysis on the cVEDA cohort (1,143+ subjects), investigating associations between brain structure, connectivity, white-matter microstructure, and behavioral phenotypes.
- Built unified multimodal datasets by integrating structural MRI, functional MRI, diffusion MRI, and behavioral assessments, with extensive preprocessing, quality control, and age-residualization pipelines.
- Designed machine learning and statistical modeling frameworks using ElasticNet, Random Forest, XGBoost, Gradient Boosting, ANOVA, mRMR, and ReliefF to discover stable neurobiological biomarkers.
- Identified shared and phenotype-specific neural signatures across adversity, substance-use, and behavioral domains through dimensionality reduction, multimodal feature analysis, and interpretable brain-region ranking.

Escorts Kubota Finance Ltd
Information Technology Intern
December 2024 – January 2025
Faridabad, India

- Collaborated with IT, Compliance, Credit, and Business Development teams to analyze NBFC operations, lending lifecycles, credit underwriting workflows, portfolio diversification, and business strategy execution.
- Assessed enterprise technology systems including the iBPS framework and Azure-based architecture, evaluating workflow automation, scalability, operational resilience, and regulatory compliance mechanisms.
- Examined RBI-mandated controls and risk evaluation frameworks, identifying automation and data-integration opportunities to streamline manual processes, improve reporting efficiency, and strengthen cross-functional coordination.

Apollo Tyres Ltd
Data Engineering Intern
May 2024 – July 2024
Gurgaon, India

- Developed AWS Lambda functions to automate end-to-end data processing workflows from S3 to Redshift.
- Processed and transformed raw data files (log, CSV, Excel) using Python (Pandas) to ensure seamless integration and data accuracy.
- Designed and maintained Redshift schemas and tables using DBeaver for efficient queries and reporting.
- Implemented advanced data handling techniques, including pivoting, multi-index manipulation, validation, and batch insertions to optimize performance.
- Leveraged AWS S3 triggers and events to orchestrate automated workflows for data ingestion and processing.
- Developed production-ready AWS data pipelines used by the R&D team for data processing and analysis.

The Modern Data Company
Product Design Intern
July 2023 – August 2023
Indore, India

- Collaborated with the Product team to redesign the product and homepage, improving user experience.
- Conducted in-depth analysis to identify design gaps and created detailed wireframes using Whimsical.
- Developed high-fidelity prototypes on Figma, adhering to industry best practices and standards.
- Enhanced UI/UX, tailoring improvements to meet specific client needs and boost user satisfaction.

PROJECTS

InvestShield – SEBI National Securities Market Hackathon Award Winner
AI-powered Fraud Detection Platform
2025

- Developed an award-winning machine learning platform for detecting fraudulent investment content, scam announcements, and fake advisor claims across social media and online sources.
- Collected and processed 50,000+ verified samples from Telegram, WhatsApp, websites, and regulatory sources; built data cleaning, labeling, and feature extraction pipelines across 7 fraud categories.
- Engineered an ensemble ML architecture using Random Forest, Gradient Boosting, and Logistic Regression with soft voting to achieve 95%+ overall detection accuracy.
- Built a full-stack FastAPI–React platform for real-time fraud assessment with sub-second latency and 100+ concurrent-user scalability, integrating REST APIs, Docker, and modular microservices.
- Recognized among 120+ national teams for innovation, technical feasibility, and direct alignment with SEBI's investor protection and market supervision mandates.

LEADERSHIP & EXTRACURRICULAR

- **Fundraising Intern, FHLE** (Dec 2024 – Present) – Raised 50,000+ for health and learning programs; engaged donors and promoted awareness campaigns.
- **Marketing Intern, Youth India Foundation** (Sep – Dec 2024) – Designed youth-centric content and digital campaigns promoting sustainable entrepreneurship; offered leadership position of President.
- **Campus Ambassador, Kratos Gaming Network** (Feb – Jun 2023) – Hosted Web3 gaming sessions and competitions; expanded on-campus player community for P.O.G. Engine.
- **Core Member, Anubhava Committee** (Jun 2023 – Present) – Conducted 15+ student and alumni interviews; produced storytelling content on Medium and YouTube.
- **Editorial Lead, Zense Club** (Aug 2023 – Aug 2024) – Led editorial design for IIITB's software developers' club; authored newsletters, event summaries, and tech spotlights.
- **Core Member, Chhayachitra** (Sep 2023 – Sep 2024) – Photographed flagship campus fests, research symposiums, and social initiatives; contributed to IIITB's visual media archive.
- **Core Member, Sports Committee** (Aug 2023 – May 2024) – Organized intra- and inter-college tournaments; oversaw logistics, scheduling, and coordination with sports captains.